

Identification of an Unknown Sulfur Compound Using XANES and Linear-Combination Fitting

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Abstract

To identify the chemical environment, oxidation state, and bonding of an element—factors that directly affect how a material behaves and reacts—simply confirming the presence of an element is insufficient, because it cannot reveal how the element is bonded or what form it takes; XANES addresses this issue, as spectral features near the absorption edge change with the electronic and chemical environment of the element, and these changes can be studied through the position and shape of the features. For sulfur, sulfur K-edge XANES can distinguish different sulfur species even when they all contain sulfur. In response to the challenge of identifying an unknown sulfur-containing compound, this paper proposes a method based on sulfur K-edge XANES, which is sensitive to the electronic structure and local chemical environment of sulfur and can therefore be used to distinguish different sulfur compounds. This study combines experimental XANES data processing with reference-spectrum fitting: after background subtraction and normalization, the unknown spectrum was compared with the reference spectra of dibenzothiophene (DBT), dibenzyl sulfide, sulfoxide, and sulfone, and linear-combination fitting was performed using Excel Solver to determine the reference combination that best reproduced the unknown spectrum. The results show that the unknown sample is most consistent with dibenzothiophene (DBT) among the reference compounds tested. This study demonstrates that sulfur K-edge XANES can be used to identify chemicals in an unknown sample by their chemical environment.

Keywords

XANES, sulfur K-edge, dibenzothiophene, linear-combination fitting

1. Introduction

1.1 X-ray Absorption Spectroscopy and Chemical Structure

X-ray absorption spectroscopy (XAS) measures how a material absorbs X-rays at different photon energies. As the photon energy approaches the binding energy of a core electron, the absorption coefficient increases rapidly. This sharp increase produces an absorption edge. X-ray absorption spectroscopy (XAS) measures how a sample absorbs X-rays as the X-ray energy changes. The region near the absorption edge is called X-ray absorption near-edge structure (XANES). This region provides information about the electronic state and chemical environment of the absorbing atom. The local environment around an atom can affect the features of the XANES spectrum.

Sulfur can exist in different chemical forms. The sulfur atom can have different bonds, oxidation states, and molecular structures. These differences can change the electronic structure of sulfur. They can also change how sulfur absorbs X-rays. Sulfur K-edge XAS can detect these changes because X-ray absorption depends on the chemical environment around the sulfur atom [1]. Paris et al. studied how sulfur K-edge absorption changes with oxidation state [3]. They found that sulfur compounds with different oxidation states can show different K-edge spectra.

1.2 Why Sulfur K-edge XANES Can Distinguish Sulfur Compounds

Sulfur K-edge XANES can show changes in the chemical environment around sulfur. Sulfur compounds can have different structures and oxidation states. These differences can change the electronic states of sulfur. They can also change the position and shape of features near the absorption edge. Mijovilovich et al. studied sulfur K-edge XANES spectra from several model compounds. The compounds included dibenzothiophene and oxidized sulfur compounds. Their results showed that compounds with different electronic structures can have similar XANES features [2]. This finding means that one peak may not be enough to identify an unknown sulfur compound. Reference spectra can help with the identification of an unknown sample. Researchers can compare the unknown spectrum with spectra from compounds with known sulfur environments. The comparison can include the overall shape of the spectrum. It does not need to focus on only one peak. Waldo et al. studied sulfur K-edge XANES features from different sulfur forms. Their work included thiophenic sulfur, oxidized sulfur, and sulfate species [6].

1.3 Linear-Combination Fitting of Sulfur XANES

Linear-combination fitting (LCF) provides a systematic approach to interpreting unknown XANES spectra. In this method, the measured spectrum is modeled as a sum of reference spectra:

$$\mu_{unknown}(E) \approx \sum_{i=1}^n c_i \mu_i(E) \quad (1)$$

Where $\mu_{unknown}(E)$ represents the unknown spectrum, $\mu_i(E)$ is the spectrum of reference compound i , and c_i is the fitted coefficient for that component.

Researchers have used LCF to analyze sulfur species with sulfur K-edge XANES. Prietzel et al. compared LCF with spectral deconvolution to identify different sulfur forms in soil. Their results showed that LCF can provide useful estimates when the reference compounds are well chosen. However, the results also depend on the composition of the sample and how well the reference spectra represent the sulfur environments in that sample [4]. This issue also matters in the present study because the choice of reference spectra can affect the identification of the unknown sample.

1.4 Present Study

The main research question of this study is whether synchrotron-based sulfur K-edge XANES can be used to identify the chemical environment of sulfur in an unknown sample. The study also examines whether direct comparison with reference spectra and linear-combination fitting can identify the unknown sample. In particular, the analysis tests which reference compound or combination of compounds gives the closest match to the unknown spectrum. The dataset was collected as part of the Pioneer Research Program. It contains one unknown sulfur-containing sample and reference spectra from four compounds: dibenzothiophene (DBT), dibenzyl sulfide, sulfoxide, and sulfone. The spectra were first processed using background subtraction and normalization. These steps were completed before the spectra were compared or fitted. The study then used Excel Solver to perform linear-combination fitting. The fitting process tested different combinations of the reference spectra and found the combination that most closely matched the unknown spectrum. The Pioneer analysis identified the unknown sample as DBT. This result provides a useful way to test sulfur K-edge XANES for sample identification. The analysis begins with the basic principles of synchrotron radiation and X-ray absorption. The study then examines the main features of the XANES spectrum and processes the data

for comparison. The unknown spectrum is compared with spectra from known reference samples. Linear-combination fitting is then used to measure how well the reference spectra match the unknown sample.

2. Methodology

2.1 XANES Dataset and Reference Spectra

This study uses sulfur K-edge XANES data from the Pioneer Research Program. The dataset includes one unknown sample and reference spectra from several compounds. The compounds have different sulfur environments. The analysis used four reference compounds: dibenzothiophene (DBT), dibenzyl sulfide (DBS), sulfoxide, and sulfone. Each reference compound represents a different chemical environment. Their spectra were used as reference data for comparison with the unknown sample. The analysis compared the overall spectral features of the reference compounds with those of the unknown spectrum. The analysis have performed in several step. First, processing the data by removing the background and normalization. Second, Aligning the energy grid with all reference data by equalize them. Finally, using excel solver to perform linear combination fitting to help find the coefficients.

2.2 Data Processing

Direct comparison with the raw data will not help us find the result, because background and intensity variations will create differences even within the same data. Therefore, background subtraction and normalization were applied before fitting to help remove the effects of background and intensity variations. Here is the equation used for subtracting the background and normalization:

$$\mu_{sub}(E) = \mu_{raw}(E) - \mu_{bg}(E) \quad (2)$$

$$\mu_{norm}(E) = \frac{\mu_{sub}(E) - \mu_{pre}}{\mu_{post} - \mu_{pre}} \quad (3)$$

$$\mu_{raw}(E) = \text{rawdata} \quad (4)$$

$$\mu_{sub}(E) = \text{background} - \text{subtracted absorption} \quad (5)$$

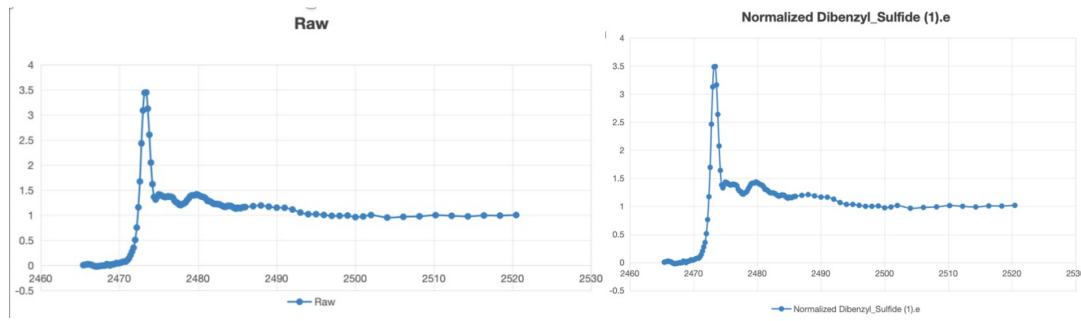
$$\mu_{pre} = \text{pre_edge absorption level} \quad (6)$$

$$\mu_{post} = \text{post_edge absorption level} \quad (7)$$

$$\mu_{norm}(E) = \text{normalized absorption} \quad (8)$$

A linear pre-edge baseline was established over 2465.4–2469.4 eV, and normalization was performed using the post-edge region 2500–2520 eV, with an edge energy E_0 of 2472 eV. These preprocessing steps were applied consistently to both reference and unknown spectra. As shown in Figure 1, the normalization procedure modifies the spectral profile of dibenzyl sulfide.

Figure 1: Example showing the effect of normalization on the data of dibenzyl sulfide



2.3 Energy-Grid Alignment

The reference spectra and the unknown spectrum were measured at different energy points, so their energy grids were not directly aligned. To enable comparison and linear-combination fitting, we need to equalize the data; to do that, we plug in this equation:

$$\mu(E) = \mu_1 + \frac{\mu_2 - \mu_1}{E_2 - E_1}(E - E_1) \quad (9)$$

where E = the energy point we want to estimate. E_1, E_2 = the two surrounding measured energy points. μ_1, μ_2 = the absorption values at E_1 and E_2 . $\mu(E)$ = the estimated absorption value at E .

2.4 Linear-Combination Fitting

The unknown spectrum was modeled as a linear combination of the reference spectra. In this study specifically, we use this equation:

$$Unknown = A(DBT) + B(DBS) + C(Sulfoxide) + D(Sulfone) \quad (10)$$

which ABCD corresponds to the percentage of each compound, and their sum has to be equal to one.

I used Solver in Excel to determine the optimal fitting coefficients. In this process, physical constraints are necessary: all coefficients were constrained to be non-negative and less than 1 ($1 \geq A, B, C, D \geq 0$) and to sum to unity ($A + B + C + D = 1$), so that they represent relative contributions of each reference compound rather than unrestricted fitting parameters. As shown in Figure 2, cells \$M\$2:\$M\$5 were defined as variable cells for fitting coefficients, and cell \$M\$6 was set to satisfy the constraint that the sum of coefficients equals 1. The objective cell \$H\$301 was minimized using the GRG Nonlinear solving method.

Figure 2: Parameter configuration interface of Excel Solver for linear-combination fitting

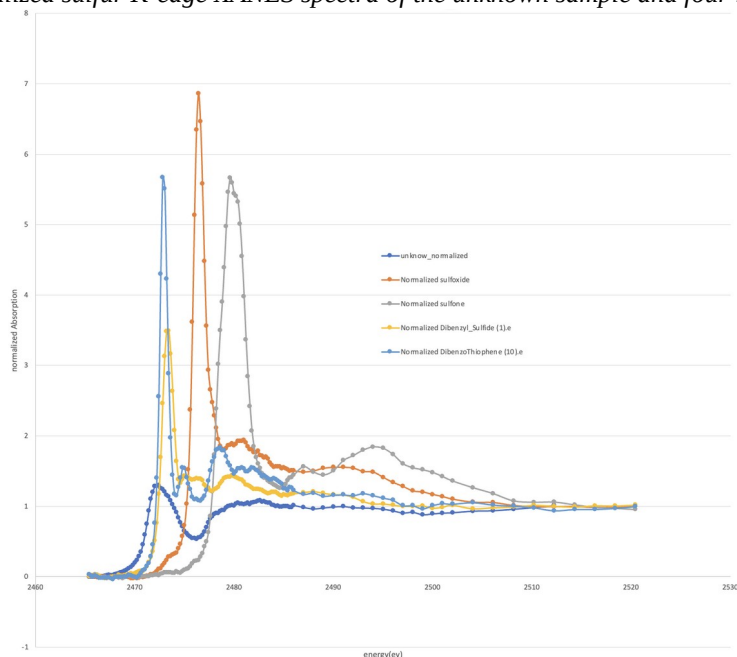
The image shows the 'Solver Parameters' dialog box in Microsoft Excel. The 'Set Objective' field is set to '\$H\$301'. The 'To' section has 'Min' selected, and the 'Value Of' field is set to '0'. The 'By Changing Variable Cells' field is set to '\$M\$2:\$M\$5'. The 'Subject to the Constraints' list contains three constraints: '\$M\$2:\$M\$5 <= 1', '\$M\$2:\$M\$5 >= 0', and '\$M\$6 = 1'. To the right of these constraints are buttons for 'Add', 'Change', 'Delete', 'Reset All', and 'Load/Save'. Below the constraints, there is a checkbox for 'Make Unconstrained Variables Non-Negative' which is currently unchecked. The 'Select a Solving Method' dropdown is set to 'GRG Nonlinear', with an 'Options' button next to it. A 'Solving Method' text box provides instructions: 'Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.' At the bottom are 'Close' and 'Solve' buttons.

3. Results

3.1 Spectral features

The unknown sample exhibits its main absorption maximum at 2472.79 eV within the S K-edge region. As shown in Figure 3, when compared to reference compounds, this peak position is much closer to dibenzothiophene (~2474.00 eV) than to the other references, such as sulfoxide (~2476.40 eV) and sulfone (~2479.60 eV). Combined with a visual comparison of the overall spectral shape, the unknown spectrum resembles dibenzothiophene, providing an initial basis to identify the unknown sample as DBT.

Figure 3: Normalized sulfur K-edge XANES spectra of the unknown sample and four reference compounds



3.2 LCF Results

The analysis adopted linear-combination fitting using four reference spectra: dibenzothiophene (DBT), dibenzyl sulfide (DBS), sulfoxide, and sulfone. The fitting was performed over the energy range of 2469–2490 eV, and the fitted model reproduces the measured spectrum of the unknown sample well within this interval. As shown in Table 1, the coefficients solved by Excel Solver show that dibenzothiophene accounts for 0.8942 and sulfone accounts for 0.1058, while the coefficients of the other two reference compounds equal zero. DBT exhibits the largest fitted coefficient among the four reference spectra, whereas sulfoxide and dibenzyl sulfide contribute negligibly to the fitted model. These fitted coefficients quantify the relative contribution of each reference spectrum to the reconstructed spectral model; a larger coefficient corresponds to a higher weight of the reference spectrum. It should be noted that these coefficients cannot be directly interpreted as molar concentrations of compounds in the original sample. Instead, they only describe the relative spectral contribution of each reference compound within the current fitting framework.

Table 1: Fitted coefficients obtained from linear-combination fitting of sulfur K-edge XANES spectra

Sulfoxide	0
Sulfone	0.10582364
Dibenzothiophene	0.89417636
Dibenzyl sulfide	0
	1

4. Discussion

4.1 Interpretation of the XANES Spectrum

A direct comparison with one reference spectrum cannot fully identify the unknown sample. Mijovilovich et al. showed that compounds with different molecular electronic structures can have similar XANES features [2]. A close spectral match can show that the unknown sample and a reference compound have similar sulfur environments. The match can support the comparison, but it does not show that the two samples are the same molecule. The reference spectra used in this study showed clear differences near the absorption edge as the sulfur environment changed. The lower-energy features of reduced and thiophenic sulfur were different from the features of more oxidized sulfur, such as sulfone. These differences are consistent with earlier work showing that sulfur oxidation state and bonding can affect the energy position and shape of the absorption edge [1]. In this study, the unknown spectrum showed the closest overall match to the DBT reference. Mijovilovich et al. also used DBT as a model compound for thiophenic sulfur and reported characteristic features in its sulfur K-edge spectrum that are related to its electronic environment [2]. Based on the spectral comparison, the sulfur in the unknown sample is therefore consistent with a thiophenic sulfur environment and is most consistent with DBT among the reference compounds tested.

4.2 Role of Linear-Combination Fitting

LCF provides a more consistent way to compare the unknown spectrum with several reference spectra at the same time. The method reduces the difference between the measured spectrum and a model made from known reference spectra. This gives a numerical comparison in addition to visual inspection. Prietzel et al. used LCF with sulfur K-edge XANES to study different sulfur forms [4]. In the present study, DBT had the largest coefficient among the four reference spectra. The result supports the DBT identification because the reference with the largest contribution also showed the closest match to the unknown spectrum. The fitted coefficient should be interpreted carefully. A large coefficient means that the reference spectrum has a larger effect on the shape of the fitted spectrum. The fitted coefficient does not show that the original sample contains the related compound at the same percentage. Several factors can affect the fitted value. The reference spectra can affect the result because differences between them can change the fit. The normalization method can also

change the coefficient values. Experimental noise can affect the fitting result as well. The fitting model also depends on certain assumptions, which can lead to changes in the fitted values.

4.3 Influence of Data Processing

XANES data needs to be processed before the spectra are compared and fitted. Background subtraction removes unwanted background signals from the measured spectrum. Normalization then puts the spectra on a similar scale. These steps make it easier to compare an unknown spectrum with reference spectra. The processing method can affect the analysis. Different background subtraction methods can change the shape of a spectrum. Different normalization methods can also change the spectrum. These changes can affect the fitted coefficient values [5]. The LCF coefficients should therefore be treated as contributions to the fitted spectrum rather than direct measurements of chemical concentration. Clear reporting of the preprocessing method is also important because it allows the analysis to be repeated and the fitting results to be evaluated. This information allows other researchers to repeat the analysis and better judge the reliability of the fitting results.

5. Conclusion

This study used synchrotron-based sulfur K-edge XANES to examine the chemical environment of sulfur in an unknown sample. The analysis connected the basic principles of X-ray absorption with the processing and interpretation of experimental XANES data. After background subtraction and normalization, the unknown spectrum was compared with reference spectra from different sulfur environments. LCF was then used to compare the measured spectrum with a spectrum reconstructed from the reference compounds. The analysis gave the closest overall match to dibenzothiophene (DBT). DBT also had the largest coefficient in the LCF model. These two results support the assignment of the unknown as being consistent with DBT within the reference set used in this study. This result is also consistent with previous work that has used DBT as a model compound for thiophenic sulfur. The LCF coefficients should not be treated as direct chemical concentrations. The coefficients depend on factors such as the reference spectra and the preprocessing steps. A spectral match cannot prove the molecular identity of an unknown sample.

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Conflicts of Interest

The authors declare no conflict of interest.

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